Formal Language & Automata Theory_2023_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: [1 x 10 = 10]

1. NFA, in its name has 'non-deterministic' because of _____
2. The non-Kleene Star operation accepts the following string of finite length over set A = {0,1} | where string s contains even number of 0 and 1
3. Language of finite automata is of which type?
4. The concept of FSA is much used in _____ part of the compiler
5. FSM can recognize _____
6. Consider the following language: L = {anbn|n = 1}; L is ___
7. Set of regular languages over a given alphabet set is closed under _____
8. Consider the grammar:

   S → ABCc | Abc;

   BA → AB;

   Bb → bb;

   Ab → ab;

   Aa → aa.

   Write the sentences can be derived by this grammar?
9. Consider the following grammar:

   S →Ax / By;

   A→ By/Cw;

   B → x / Bw;

   C→y

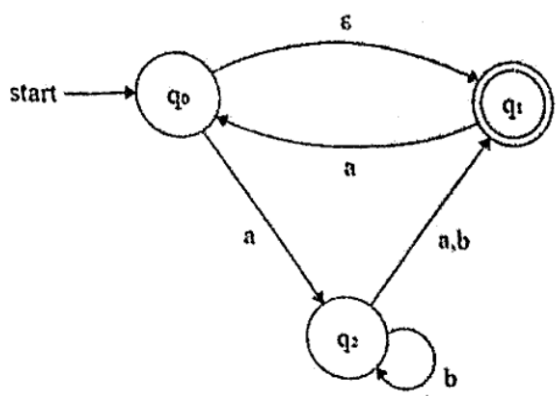
   Write the regular expressions describe the same set of strings as the grammar.
10. Let S = {a, b, c, d, e}. The number of strings is _____ in S* of length 4 such that no symbol is used more than once in a string.
11. Given a grammar G, a production of G with a dot at some position of the right side is called _____.

12. Number of states of the FSM required to simulate behaviour of a computer with a memory capable of storing 'm' words, each of length 'n' is ___.

Group-B (Short Answer Type Question)

Answer any three of the following: [5 x 3 = 15]

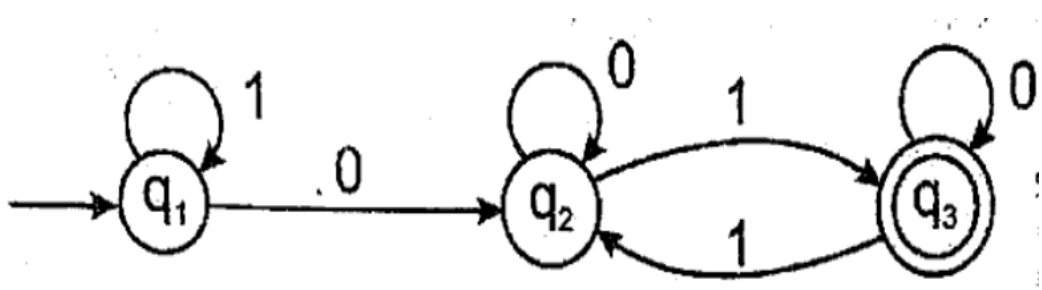
2. Design a DFA where every string either starts with 01 or ends with 01 over the alphabet set {0,1}. [5]
3. Write the regular expression for the language $L = a^n | n > 0$. [5]
4. Construct an NFA for the regular expression (0+1)*00(0+1)*. [5]
5. Design a PDA for the language $L = WcW^R | \, w \in a, b^*$. [5]
6. Convert the following NFA to DFA. [5]



Group-C (Long Answer Type Question)

Answer any three of the following: [15 x 3 = 45]

7. (a) Design a DFA where each and every string end with '001' over the alphabet set {0,1}. [5]
   (b) Obtain the regular expression for the following DFA. [5]



(c) Consider the following e-NFA:

Compute the e-closure of each state. Convert the NFA to DFA. [5]

δ	ϵ	a	b
$\rightarrow p$	$\{r\}$	$\{q\}$	$\{p, r\}$
q	$\emptyset$	$\{p\}$	$\emptyset$
$* \ r$	$\{p, q\}$	$\{r\}$	$\{p\}$

8. (a) Define Chomsky normal form and convert the following CFG to CNF:

$S \rightarrow aSb|ab|Ab, A \rightarrow aab$ [6]

(b) What is useless production? Eliminate ε, unit and useless production from following grammar:

$$A \rightarrow bA|Bba|aa,\ B \rightarrow aba|b|D,\ C \rightarrow CA|AC|B,\ D \rightarrow a|\varepsilon$$ [9]

9. (a) Define Deterministic PDA and Non-deterministic PDA. [6]

(b) Construct a PDA for the grammar $S \rightarrow aAA, A \rightarrow aS|bS|a$ [9]

10. (a) State the Pumping lemma for the Regular Language (RL). [4]

(b) State the Pumping lemma for the Context Free Language (CFL). [4]

(c) Prove that the given language is not regular: $L = a^n b^n \geq 0$ [7]

11. Transform the CFG into GNF, given $G = (A_1, A_2, A_3, a, b, P, A_1)$ and production P as:

$A_1 \rightarrow A_2A_3,\ \ A_2 \rightarrow A_3A_1|b,\ \ A_3 \rightarrow A_1A_2|a$ [15]